Contrastive Multiview Subspace Clustering of Hyperspectral Images Based on Graph Convolutional Networks

Renxiang Guan, Zihao Li, Wenxuan Tu, Jun Wang, Yue Liu, Xianju Li, *Member, IEEE*, Chang Tang, *Senior Member, IEEE*, and Ruyi Feng, *Member, IEEE*

***Abstract*— High-dimensional and complex spectral structures make the clustering of hyperspectral images (HSIs) a challenging task. Subspace clustering is an effective approach for addressing this problem. However, current subspace clustering algorithms are primarily designed for a single view and do not fully exploit the spatial or textural feature information in HSI. In this study, contrastive multiview subspace clustering of HSI was proposed based on graph convolutional networks. Pixel neighbor textural and spatial–spectral information was sent to construct two graph convolutional subspaces to learn their affinity matrices. To maximize the interaction between different views, a contrastive learning algorithm was introduced to promote the consistency of positive samples and assist the model in extracting robust features. An attention-based fusion module was used to adaptively integrate these affinity matrices, constructing a more discriminative affinity matrix. The model was evaluated using four popular HSI datasets: Indian Pines, Pavia University, Houston, and Xu Zhou. It achieved overall accuracies of 97.61%, 96.69%, 87.21%, and 97.65%, respectively, and significantly outperformed state-of-the-art clustering methods. In conclusion, the proposed model effectively improves the clustering accuracy of HSI. Our implementation is available at https://github.com/GuanRX/CMSCGC.**

***Index Terms*— Contrastive learning, hyperspectral images (HSIs), multiview clustering, remote sensing, subspace clustering.**

NOMENCLATURE

n	Number of data points.
p	Number of views.
w	Input size of the model.
k	Number of neighbors selected for each node.
d^p	Dimension of X^p.
$\hat{d}^p$	Dimension of Z^p.
$X^p \in \mathbb{R}^{d^p \times n}$	Data matrix on the pth view.
$Z^p \in \mathbb{R}^{\hat{d}^p \times n}$	Feature matrix on the pth view.
$C^p \in \mathbb{R}^{n \times n}$	Self-expression matrix learned on the pth view.
$A^p \in \mathbb{R}^{n \times n}$	Adjacency matrices on the pth view.
$Y^p \in \mathbb{R}^{n \times n}$	Affinity matrices on the pth view.

Manuscript received 14 November 2023; revised 7 January 2024 and 30 January 2024; accepted 22 February 2024. Date of publication 26 February 2024; date of current version 13 March 2024. This work was supported in part by the Natural Science Foundation of China under Grant 42071430 and in part by the Opening Fund of Key Laboratory of Geological Survey and Evaluation of Ministry of Education under Grant GLAB2022ZR02 and Grant GLAB2020ZR14. An earlier version of this paper was presented at APWEB-WAIM 2023. *(Corresponding author: Xianju Li.)*

Renxiang Guan is with the School of Computer Science, China University of Geosciences, Wuhan 430074, China, and also with the College of Computer, National University of Defense Technology, Changsha 410073, China (e-mail: renxiangguan@nudt.edu.cn).

Zihao Li, Xianju Li, Chang Tang, and Ruyi Feng are with the School of Computer Science, China University of Geosciences, Wuhan 430074, China (e-mail: lizihao@cug.edu.cn; ddwhlxj@cug.edu.cn; tangchang@cug.edu.cn; fengry@cug.edu.cn).

Wenxuan Tu, Jun Wang, and Yue Liu are with the College of Computer, National University of Defense Technology, Changsha 410073, China (e-mail: wenxuantu@163.com).

Digital Object Identifier 10.1109/TGRS.2024.3370633

I. INTRODUCTION

ADVANCEMENTS in spectral imaging technology have led to the emergence of hyperspectral images (HSIs) as potent instruments for detection methodologies, fostering progress in domains such as environmental surveillance [1], national defense and security [2], and mineral discernment [3], [4]. Unlike conventional color imagery, HSI exhibits enhanced resolution and a wealth of spectral data, providing a more precise representation of terrestrial information [5], [6]. This has led to the development of an array of HSI processing techniques [7].

In recent decades, considerable advancements have been made in supervised HSI classification techniques [8], [9], encompassing machine learning models such as support vector machines [10] and deep learning models such as transformer and convolutional neural networks (CNNs) [11]. These methods are predicated on the necessity of manually labeling data for training samples, a process that is both labor-intensive and reliant on domain expertise [12]. To alleviate the human workload and address the limited availability of labeled information in HSI, unsupervised learning approaches, such as clustering, have garnered significant attention [13], [14]. Clustering of HSI enables automated data processing and interpretation; however, owing to the substantial spectral variability and intricate spatial structures of HSI, clustering poses a formidable challenge [15].

HSI clustering entails the classification of pixels into corresponding groups and the segmentation of pixels into distinct categories based on their inherent similarity, maximizing intraclass and minimizing interclass similarities [16]. Numerous clustering algorithms have been proposed, including those centered on cluster centroids, such as k-means [17] and fuzzy c-means clustering [18], as well as those predicated on the spatial density distribution of features, exemplified by the

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:41:11 UTC from IEEE Xplore. Restrictions apply.

mean shift algorithm [19], and the clustering approach based on ensemble density analysis [20]. Nonetheless, these methods are sensitive to initialization and noise, with an overreliance on similarity measures. To further explore the underlying structure of HSI data, subspace clustering algorithms [21], [22], [23], [24] have attracted considerable interest with noteworthy results.

Subspace clustering algorithms effectively integrate traditional feature selection techniques with clustering approaches to process feature subsets or weights corresponding to each data cluster during the partitioning of samples [25]. Notable examples include sparse subspace clustering (SSC) [26] and low-rank subspace clustering [27], which rely on identifying the sparse representation matrix of the original data, constructing a similarity graph on the corresponding matrix, and employing spectral clustering to derive clustering outcomes. Recognizing the wealth of spatial environmental information conveyed by the HSIs as a data cube, Zhang et al. [26] introduced an SSC method to enhance clustering efficacy. In [28], the l_2-norm regularized SSC (l_2-SSC) algorithm incorporated adjacent information into the coefficient matrix through l_2-norm regularization constraints. In addition, Huang et al. [29] demonstrated the improved utilization of spatial information in localized regions via semi-supervised joint constraints, augmenting the clustering performance of the algorithm.

Although traditional subspace-clustering methods have yielded satisfactory results, their robustness and accuracy remain limited in complex HSI scenarios. Deep clustering models address these shortcomings by extracting profound and robust features [16], [30], [31], [32]. Lei et al. [33] introduced a deep model that employs multilayer autoencoders for self-expression learning. Li et al. [34] integrated self-supervised subspace clustering with an adaptive initialization for HSI and accomplished significant results. To extract robust nonlinear affinity, Cai et al. [35] proposed a graph regularized residual subspace clustering network (GR-RSCNet), which employs a deep neural network architecture combined with graph regularization for HSI subspace clustering. Moreover, subspace clustering models based on graph convolutional networks (GCNs) have gained popularity because of their capacity to capture neighborhood information. Cai et al. [36] effectively integrated structural and feature information from a graph representation learning standpoint and proposed a graph convolutional subspace clustering (GCSC) method. Zhang et al. [37] and Huang et al. [38] utilized hypergraph convolutional subspace clustering to thoroughly exploit high-order relationships and extensive interdependencies.

Despite improvements in clustering performance, the aforementioned algorithms still have two primary limitations. First, directly applying these methods to HSIs often generates cluster maps with considerable noise due to constraints in discriminative information within the spectral domain, the intricacy of ground objects, and the heterogeneity of spectral features within the same class [39]. Second, these methods are predominantly tested on a single view, and an abundance of experimental evidence suggests that incorporating complementary information from multiple views can effectively enhance clustering accuracy [40], [41], [42]. Tian et al. [43] conducted multiview clustering, although it exhibited noise sensitivity. Chen et al. [44] employed multiview subspace clustering on a polarized HSI, providing a sparse representation of all data independently by preclustering each view. Huang et al. [38] integrated local and nonlocal spatial information to discern a shared intrinsic cluster structure, improving the overall clustering performance. Lu et al. [45] merged spectral and spatial information to establish capability regions using multiview kernels for collaborative subspace clustering to achieve superior clustering outcomes. The above method shows that multiview complementary information can improve the accuracy of clustering tasks; however, it is limited to the use of multiview attributes or multiple branch networks. Not only is there no assistance from other view information in the feature-learning stage, but the information weights of distinct views are neglected, potentially leading to the loss of vital information.

To address these challenges, we introduced a novel multiview subspace clustering approach for HSIs that effectively amalgamates textural information with spectral–spatial information. By incorporating a GCN and contrastive learning methodology [46], [47], we aggregated neighborhood information while leveraging the contrastive learning technique to maximize the consistency between views during representation learning. Finally, a more discriminative matrix was obtained using adaptive fusion affinity matrices based on the attention fusion module. The principal components of this study are as follows.

1) We proposed a novel deep multiview subspace clustering algorithm for HSI clustering, denoted as CMSCGC, capable of concurrently learning textural and spectral–spatial information.
2) The integration of contrastive learning within the CMSCGC maximized the consistent representation of views, enabling the model to extract more robust feature representations.
3) An attention-based fusion module was used to adaptively harness the affinity matrices, constructing a more discriminative affinity matrix.
4) We systematically evaluated our method on four highly competitive datasets, demonstrating that our proposed method significantly outperforms state-of-the-art techniques.

The structure of the rest of this article is outlined as follows. The brief reviews of related works on multiview subspace clustering, HSI clustering, and contrastive learning are provided in Section II. Section III presents the details of the proposed CMSCGC method. The various results are reported and analyzed in Section IV, followed by a concluding summary in Section V.

II. Related Work

In this section, we present an overview of multiview subspace clustering and its foundational models, followed by a concise review of the developments of HSI clustering and contrastive learning.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:41:11 UTC from IEEE Xplore. Restrictions apply.

A. Multiview Subspace Clustering

Multiview clustering approaches [48], [49], [50], [51] have achieved considerable success across various domains. The field of remote sensing images [38] can also benefit from multiview complementary information to facilitate clustering tasks. Given multiview data $\{X^p\}_{p=1}^P$, where $X^p \in \mathbb{R}^{d^p \times n}$ denotes the pth view data with an upper dimension of d^p and n is the number of samples, subspace clustering postulates that every point can be expressed by a linear combination of other points within the same subspace. Based on this assumption, data X^p for each view were employed as a dictionary to construct the subspace representation model as follows:

$$X^p = X^p C^p + E^p \tag{1}$$

where $C^p \in \mathbb{R}^{n \times n}$ and E^p represent the self-expression matrix for each view and representation error, respectively. Multiview subspace clustering methods are typically described as follows:

$$\begin{aligned} &\min_{C^p} \ \left\|X^p - X^p C^p\right\|_F^2 + \lambda f\left(C^p\right) \\ &\text{s.t.} \ \ C^p \geq 0, C^{p\top}\mathbf{1} = \mathbf{1} \end{aligned} \tag{2}$$

where $C^{p\top}\mathbf{1} = \mathbf{1}$ ensures that the sum of each column in C^p is 1, indicating that every sample point can be reconstructed through a linear combination of the other samples. $C_{i,j}^p$ represents the weight of the edge between the ith and jth samples. Hence, C^p can be viewed as an $n \times n$ undirected graph and $\|\cdot\|_F$ represents applying Frobenius norm to the matrix and $f(\cdot)$ is a regularization function, and λ is a parameter that balances regularization and loss. Different $f(\cdot)$ functions yield self-expression matrices C^p that satisfy various constraints. For instance, sparse constraints have been applied to matrices [26] or low-rank representations of data [27].

Upon obtaining self-expression matrices $\{C^p\}_{p=1}^P$ for each view, they are fused to create a unified self-expression matrix $C^p \in \mathbb{R}^{n \times n}$. In certain methods, this step is performed concurrently with the learning phase across all the views. Using a consistent C^p as the input for spectral clustering yields the final clustering outcome. Implementing spectral clustering first requires the construction of a similarity matrix, a step that typically involves calculating the direct similarity of all pairs of samples in the dataset, resulting in a time complexity and space complexity of $\mathcal{O}(n^2)$. Then, proceed to compute the eigenvectors of the Laplacian matrix, which has a time complexity of $\mathcal{O}(n^3)$ for dense matrices. Therefore, the overall time complexity of (2) is $\mathcal{O}(n^3)$.

B. HSI Clustering

Owing to the high dimensionality, high correlation, and complex distribution of spectral data, the analysis of hyperspectral data is challenging. Traditional supervised classification methods require a large number of labeled samples that are often difficult to obtain. In recent years, unsupervised HSI clustering has become an active research topic for overcoming this limitation. HSI clustering groups pixels with similar spectral features into different classes or clusters without prior data knowledge. Traditional clustering algorithms such as k-means are known for their simplicity and efficiency. However, their clustering results are unstable and inaccurate.

Among the existing HSI clustering methods, subspace clustering has shown promising results owing to its robustness. Representative examples include sparse and low-rank subspace clustering [26], [27]. However, these classical algorithms do not fully utilize rich information beyond spectral information. Subsequently, scholars have begun to explore the spatial information in HSIs. For instance, Zhang et al. [52] integrated spatial–spectral information into nonnegative matrix factorization and proposed a robust manifold matrix factorization (RMMF) algorithm with good performance. Cai et al. [36] introduced GCNs into hyperspectral subspace clustering, thereby enhancing the ability to combine clustering models and feature learning. Liu and Wang [53] further improved the performance of graph-subspace clustering by incorporating an optimal transport algorithm into the optimization of graph representation learning.

C. Contrastive Learning

Contrastive learning is an unsupervised technique that learns representations for specific tasks by examining the differences between similar and dissimilar samples. The primary concept is to train the model by comparing the similarities between a pair of samples, mapping them to an embedding space, and comparing their differences in this space by measuring the distances between them [54].

Contrastive learning has been widely applied in various fields, such as computer vision [55]. For example, the classical simple framework for contrastive learning (SimCLR) algorithm [56] learns latent representations by maximizing the consistency between different augmented views of the same example, demonstrating the importance of data augmentation in contrastive learning. Similarly, the momentum contrast (MoCo) for unsupervised visual representation learning algorithm [57] approaches contrastive learning from a dictionary lookup perspective while ensuring a lightweight dictionary and alleviating GPU memory constraints. Gansbeke et al. [55] generated initial clusters through feature representation learned by pretraining tasks and proposed the semantic clustering by adopting nearest neighbors (SCAN) algorithm, which explores the nearest neighbors of each image and uses this prior knowledge to train the model. Unlike the previously mentioned model, the bootstrap your own latent (BYOL) [58] model just makes the features extracted from the positive pair as similar as possible without using the negative samples. Simsiam [59] avoids model collapse by stopping the gradient. Zbontar et al. [60] proposed a model named Barlow Twins that measures the cross correlation matrix between the outputs of two identical networks fed with distorted versions of a sample, making it as close to the identity matrix as possible. Likewise, there are many novel algorithms on graph contrastive learning [61], [62] that promote graph representation learning and unsupervised learning.

Recently, contrastive learning methods have been introduced to remote sensing image tasks [63]. Hou et al. [64] employed unsupervised contrastive learning to learn robust features from HSI in preparation for subsequent classification tasks. Wang et al. [65] adopted semisupervised contrastive clustering

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:41:11 UTC from IEEE Xplore. Restrictions apply.

and selected reliable pseudo-labels for model training to detect changes in remote sensing images. Li et al. [66] proposed a global and local contrastive learning method for semantic segmentation. Wang et al. [67] integrated contrastive learning between HSIs and light detection and ranging (LiDAR) data for classification tasks.

To the best of our knowledge, existing methods have not introduced multiview contrastive learning into HSI clustering. Using contrastive learning, the consistency between the same nodes in different views or nodes from the same land-cover category can be maximized, enhancing the feature-learning capabilities and facilitating more robust clustering. The advantages of our method are demonstrated in Section IV.

III. Methodology

As shown in Fig. 1, the proposed framework consists of four crucial components: multiview graph construction, graph convolutional self-expression, contrastive learning, and attention fusion modules. First, we introduced the multiview graph construction module, followed by a description of the remaining components. The notations are summarized in the Nomenclature.

A. Multiview Graph Construction

The key objective of this module is to extract textural and space-spectral features as different views to construct a multiview graph. However, before this step, a fundamental issue must be addressed: high-dimensional HSIs contain numerous redundant bands. Therefore, we employed principal component analysis (PCA) [68] to reduce the number of bands to dimensions. Subsequently, a sliding window was used to extract the pixel and its adjacent neighboring pixels, with each patch representing its center pixel and the number of patches is the same as the number of pixels [36], [69]. In another branch, the extended morphological profile (EMP) algorithm [40] was employed to erode an image while preserving textural information. Therefore, we can obtain multiview features $\{X^p\}_{p=1}^P \in \mathbb{R}^{d^p \times n}$, where $x^p \in \mathbb{R}^{w \times w \times d^p}$ in which w and p are the patch size and the dimension of each view, respectively.

Because the GCN can only operate directly on graph-structured data, it is necessary to convert the processed data into a topological graph structure. Multiview graphs can be defined as $G^p = (V^p, E^p, X^p)$, where V^p and E^p are the corresponding node and edge sets, respectively, and the edges are denoted by the adjacency matrix A^p. Here, the data in each view are considered nodes of the graph, and the topology is constructed using their K-nearest neighbors. In each view, the adjacency matrices A^p are constructed by measuring the Euclidean distance between the centroids of different patches, which describes the global similarity of the data points. The elements in adjacency matrices A^p can be defined as

$$A_{ij}^p = \begin{cases} 1, & \mathrm{x}_j^p \in \mathcal{N}_k\left(\mathrm{x}_i^p\right) \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

where x_i^p and x_j^p are the columns of A_{ij}^p, and $\mathcal{N}_k(\mathrm{x}_i^p)$ represents the set of neighbors that include x_i in the pth view. A_{ij}^p has a value of 0 or 1, depending on whether x_i^p and x_j^p are similar.

B. Graph Convolutional Self-Expression Module

In HSI clustering tasks, learning the topological relationships between objects has long been overlooked and conventional CNN struggles to learn these relationships effectively. Thus, introducing models that can enhance object topology learning could potentially improve clustering accuracy.

Unlike models based on CNN, GCN can fully exploit the dependency relationships between nodes and the feature information of each neighboring node through graph convolution operations. Mathematically, the formula for the spectral graph convolution at the lth layer is

$$\begin{aligned} H^p(l+1) &= f\left(H^p(l), \widetilde{A}^p; W^{(l)}\right) \\ &= \sigma\left(\widetilde{D}^{p^{-\frac{1}{2}}} \widetilde{A}^p \widetilde{D}^{p^{-\frac{1}{2}}} H^p(l) W^p(l)\right). \end{aligned} \tag{4}$$

In this equation, $\widetilde{A}^p = A^p + I_N$, where I_N is the identity matrix; $\widetilde{D}^p$ and $W^p(l)$ represent the degree matrix of nodes in each view and the weight matrix of the $(l+1)$th layer, respectively; σ denotes the activation function; and $H(l)$ is for the data representation at layer l, and when l equals 0, $H^p(0)$ represents the input multiview data.

Inspired by the definition of GCSC in [36], we modified the traditional self-expression using the self-expression coefficient matrix to obtain more robust information as follows:

$$X^p = X^p \bar{A}^p C^p = Z^p C^p \text{ s.t. } \operatorname{diag}\left(C^p\right) = 0 \tag{5}$$

where C^p is the self-expression coefficient matrix for each view, and $X\bar{A}C$ can be regarded as a special linear graph convolution operation parameterized by C, where $\bar{A}$ represents the normalized adjacency matrix. The final graph self-expression is defined as

$$\min_{C^p} \frac{1}{2}\left(\lambda\left\|Z^p C^p - X^p\right\|_q + \lambda\left\|C^p\right\|_q\right) \text{ s.t. } \operatorname{diag}\left(C^p\right) = 0 \tag{6}$$

where q denotes an appropriate matrix norm and λ is the balancing coefficient. The purpose of graph convolutional self-expression is to reconstruct the original data using a self-expressive coefficient matrix C. As C includes the global structural information, a clearer dictionary can be obtained, generating a robust affinity matrix. By setting $\|\cdot\|_q$ as the Frobenius norm, the self-expression loss function can be obtained as follows:

$$\begin{aligned} \mathcal{L}_{SE} &= \frac{1}{2}\left(\left\|Z^p C^p - X^p\right\|_F + \lambda\left\|C^p\right\|_F\right) \\ &= \frac{1}{2}\mathrm{tr}\left[\left(Z^p C^p - X^p\right)^T\left(Z^p C^p - X^p\right) + \lambda C^{pT} C^p\right]. \end{aligned} \tag{7}$$

The obtained multiview self-expression coefficient matrices C^p can be used to construct affinity matrices Y^p

$$Y^p = \frac{1}{2}\left(|C^p| + |C^p|^{\mathrm{T}}\right). \tag{8}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:41:11 UTC from IEEE Xplore. Restrictions apply.

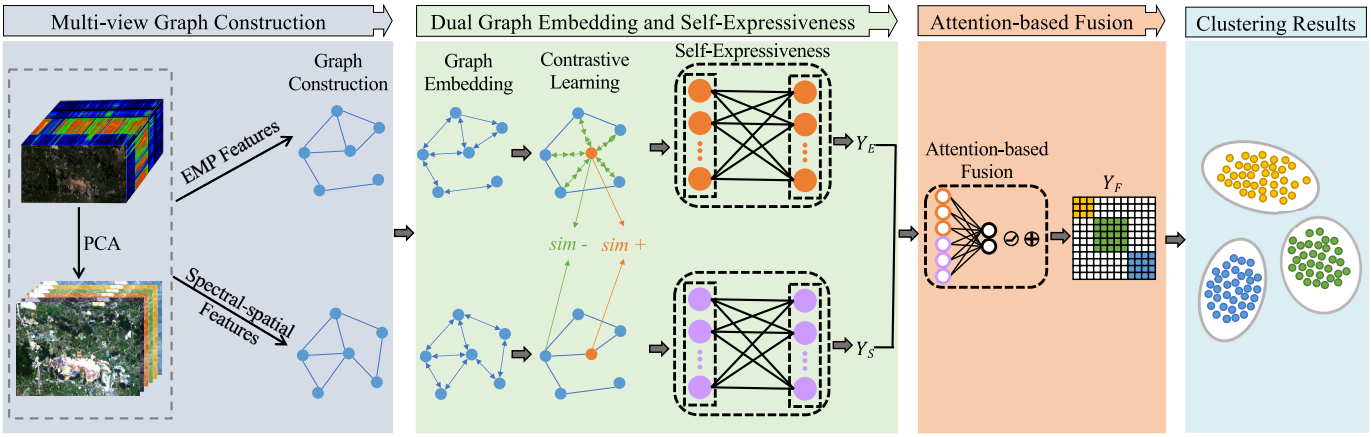


Fig. 1. Flowchart of our proposed CMSCGC framework. First, we construct multiview graphs using raw HSIs. Then, a graph CNN combined with a contrastive learning algorithm is used to learn robust features. An attention-based fusion module is then used to obtain a robust affinity matrix. Finally, spectral clustering is applied to the affinity matrix, yielding clustering results.

C. Contrastive Learning

The effective utilization of information from multiple views is a significant challenge faced by multiview clustering, that is, we must learn consistent information from different perspectives to promote better clustering. The aforementioned GCN has already extracted deep multiview features. To facilitate the subsequent steps, we represent the features of the nodes as z.

Recently, contrastive learning has demonstrated excellent performance in clustering tasks because it can maintain the desired invariance of data by constraining the distribution of similarities and dissimilarities. The concept of contrastive learning is simple: construct positive and negative samples, minimize the distance between positive samples, and remove negative samples to learn better node representations. A significant amount of work has been conducted on constructing better positive and negative samples, which play an important role in the subsequent tasks. In this study, there were naturally high-quality positive samples, that is, nodes of the same graph in different views, and random nodes from different graphs were selected as negative samples.

For any node i on the graph, we consider its node representation z_i in one view as the anchor point, and the corresponding node representations in the other views, denoted as z_i', form the positive samples. As mentioned earlier, we want z_i to be as far from the negative samples as possible and closer to the positive samples. The loss function is designed as follows:

$$\ell\left(z_i, z_i'\right) = -\log \frac{e^{\left(\text{sim}\left(z_i z_i'\right)/\tau\right)}}{\sum_{k=1}^{N} e^{\left(\text{sim}\left(z_i z_k'\right)/\tau\right)} + \sum_{k=1}^{N} e^{\left(\text{sim}\left(z_i z_k\right)/\tau\right)}} \tag{9}$$

where τ represents the temperature parameter and it is fixed to 1 in this article, sim$(\cdot)$ is the cosine similarity function, z_i' represents the negative sample representation in other views, and z_i represents the representation of the negative pair within the same layer. The overall objective for minimizing is the average of (9) for all given positive pairs, which is given by the following expression:

$$\mathcal{L}_{\text{con}} = \frac{1}{2n} \sum_{i=1}^{n} \left[\ell\left(z_i, z_i'\right) + \ell\left(z_i', z_i\right)\right]. \tag{10}$$

D. Attention Fusion Module

After obtaining affinity matrices Y^p from (8), we must fuse them to construct the final affinity graph and apply spectral clustering to the final affinity matrix. We utilized an attention-based fusion module to learn the importance a^p of each view as follows:

$$a^p = \text{att}\left(Y^p\right) \tag{11}$$

where $a^p \in R^{n\times 1}$ are used to measure the importance of each view. Details of the attention module are provided as follows. In the first step, we concatenate affinity matrices Y^p as $[Y_1, \ldots, Y_p] \in \mathbb{R}^{n\times pn}$ and introduce a weight matrix $S \in \mathbb{R}^{pn\times p}$ to capture the relationships between the self-expression matrices. Also, S is initialized with the all-ones initialization to endow the same weight to affinity matrices at the beginning of training and the updated S represents the weights of the linear layer. Next, we apply the tanh function to the product of $[Y_1, \ldots, Y_p]$ and W for a transformation. Finally, we use softmax and the l_2 function to normalize the attention values, resulting in the final weight matrix

$$a^p = \ell_2\left(\text{softmax}\left(\tanh\left(\left[Y_1, \ldots, Y_p\right] \times S\right)\right)\right). \tag{12}$$

The weight matrix can be obtained to realize the fusion operation, and the fused self-expression matrix Y_F is

$$Y_F = f\left(Y_1, \ldots, Y_p\right) = \sum_{i=1}^{N} (a_i \mathbf{1}) \odot Y_i \tag{13}$$

where $\mathbf{1} \in \mathbb{R}^{\mathbf{1}\times n}$ is a matrix with all elements being 1 and $\odot$ represents the Hadamard product of the matrix.

IV. Experiments

We demonstrate the clustering effectiveness of the proposed model on four benchmark datasets and compare them with several state-of-the-art clustering models. Some details about the experimental setup and results are presented in this section.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:41:11 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 Learning Procedure for CMSCGC

Input: HSI data $\mathbf{X}$, the input size w, the number of neighbors: k, the regularization coefficient:λ, ant the training epochs T

Output: Clustering results

```
Initialize the parameters of the network.
while epoch ≤ T do
    Feed X into Multi-view Construction module to get the multi-view data {X^p}_{p=1}^P.
    Construct graph using Eq. (3)
    Using graph convolution by to get the features Z^p Eq. (4)
    Using multi-view contrastive learning to get the deep features by Eq. (9)
    Undate parameters of the network by miniziming Eq. (10)
end while
Perform graph convolutional subspace clustering process to get the cluster labels.
```

TABLE I
STATISTICS OF THE DATASETS USED IN OUR EXPERIMENTS

Datesets	Indian Pines	Pavia University	Houston	Xu Zhou
Pixels	85×70	200×100	349×680	100×260
Abscissa	30-115	150-350	0-349	0-100
Ordinate	24-94	100-200	0-680	0-260
Channels	200	103	144	436
Samples	4391	6445	6048	12043
Clusters	4	8	12	5

TABLE II
IMPORTANT HYPERPARAMETERS SETTINGS IN OUR EXPERIMENTS

Datesets	Indian Pines	Pavia University	Houston	Xu Zhou
w	11	11	11	7
k	25	30	25	35
λ	100	1000	1000	100

A. Experiments Setup

1) Dataset: To provide a fair assessment of the model's effectiveness, it is crucial to employ a diverse dataset. This article evaluates the model's classification prowess using four widely recognized HSI datasets: Indian Pines, Pavia University, Houston-2013, and Xu Zhou, which were acquired using AVIRIS, ROSIS, ITRES CASI-1500, and HySpex sensors. For computational efficiency, we selected a subset of the scenes for our experiments. Details of the selected subscenes are listed in Table I.

a) Indian Pines: This HSI was acquired using the American AVIRIS sensor in the Indian Pines. The selected subscene is located at [30–115, 24–94], incorporating 200 spectral bands per pixel. The spatial resolution of the dataset stands at 20 m, encompassing four distinct labels.

b) Pavia University: The HSI was acquired by the German ROSIS sensor in 2003, it had a spatial resolution of 1.3 m and a spectral range of 0.43–0.86, divided into 115 bands. After removing the 12 noisy bands, the remaining 103 bands were used for clustering. The selected subscene was located at [150–350, 100–200].

c) Houston 2013: It was collected by the ITRES CASI-1500 sensor in Texas, USA, and its adjacent rural areas in 2013. After deleting the noise bands, the remaining 144 valid spectral bands are used for experiments. The Houston dataset has a spatial resolution of 2.5 mpp and contains 12 land-cover categories, such as tree, soil, water, healthy grass, running track, and tennis court. The selected subscene was located at [0–349, 0–680].

d) Xu Zhou: This HSI was located in the suburban mining area of Xuzhou City, Jiangsu Province, China. The data were obtained using an airborne HySpex imaging spectrometer with a high resolution of 0.73 m. In addition, this scene contains data for 436 bands with a spectral range of 415–2508 nm. The selected subscene was located at [0–100, 0–260] and contains five land-cover categories, including crops, bareland, red tiles, trees, and cement.

2) Evaluation Metrics: To quantitatively analyze the advantages and disadvantages of the algorithm constructed in this article and the comparison algorithm, we use three commonly used evaluation metrics to evaluate the model proposed in this article: clustering overall accuracy (OA), normalized mutual information (NMI), and Kappa coefficient (Kappa). The following introduces the evaluation metrics in detail.

1) OA is one of the most widely used clustering evaluation metrics. This refers to the proportion of correctly predicted samples among all samples. The formula for calculating the OA is as follows:

$$\mathrm{OA} = \frac{\sum_1^N \delta(y_i, \mathrm{map}(c_i))}{N} \tag{14}$$

where N is the total number of samples, y_i represents the true label of data sample x_i, and c_i represents the predicted label of the algorithm. $\delta(\cdot)$ represents the indicator function, and its value is 1 if $y_i = \mathrm{map}(c_i)$; otherwise, its value is 0. $\mathrm{map}(\cdot)$ represents the mapping function, which rearranges the obtained labels to create the best match of the true labels.

2) NMI is an index used to measure the similarity between two clustering results

$$\mathrm{NMI} = \frac{I(y_i, c_i)}{\sqrt{H(y_i)H(c_i)}} \tag{15}$$

where $I(y_i, c_i)$ represents the mutual information between true and predicted labels. The larger value of $I(y_i, c_i)$ represents the closer the relationship between y_i and c_i. $H(\cdot)$ represents the information entropy, which is the average amount of information corresponding to the entire probability distribution. The denominator in the formula is used to normalize the mutual information within the range [0, 1].

3) Kappa is defined as

$$\mathrm{Kappa} = \frac{N\sum_{i=1}^m h_{ii} - \sum_{i=1}^m (h_{i+}h_{+i})}{N^2 - \sum_{i=1}^m (h_{i+}h_{+i})} \tag{16}$$

where h_{ii} represents the count of samples correctly classified into category i, m represents the total number

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:41:11 UTC from IEEE Xplore. Restrictions apply.

TABLE III
CLUSTERING RESULTS (OA/NMI/KAPPA) OF VARIOUS METHODS ON FOUR BENCHMARK DATASETS. THE BEST AND SUBOPTIMAL RESULTS ARE SHOWN IN BOLD AND UNDERLINED, RESPECTIVELY

Dataset	Metric	k-means	CL-CNN	SSC	l_2-SSC	RMMF	NMFAML	EGCSC	GCOT	HMSC	HGCSC	GR-RSCNet	CMSCGC
InP	OA	0.5953	0.6006	0.6558	0.6645	0.7121	0.8508	0.8483	0.8404	0.8901	0.9244	0.9317	**0.9761**
	NMI	0.4483	0.3288	0.6025	0.5382	0.4985	0.7264	0.6442	0.7654	0.86	0.8921	0.8241	**0.9234**
	Kappa	0.4404	0.4057	0.5528	0.5263	0.5609	0.7809	0.6422	0.6258	0.9	0.8307	0.9008	**0.9657**
PaU	OA	0.6027	0.6283	0.6548	0.6912	0.7704	0.8967	0.8442	0.9102	0.9482	0.9531	**0.9773**	0.9669
	NMI	0.5245	0.5901	0.6706	0.6251	0.7388	0.9216	0.8401	0.8804	0.92	0.9383	0.9689	**0.9702**
	Kappa	0.6791	0.5706	0.5736	0.6881	0.6804	0.8625	0.7968	0.9233	0.83	0.9440	**0.9702**	0.9565
Hou	OA	0.3932	0.4045	0.4581	0.5097	0.7309	0.6346	0.6238	0.6584	—	0.7315	0.8399	**0.8721**
	NMI	0.4101	0.3536	0.4961	0.5816	0.7797	0.7959	0.7754	0.7729	—	0.7004	0.8731	**0.9168**
	Kappa	0.3289	0.4281	0.4013	0.4811	0.6111	0.5917	0.5812	0.6167	—	0.8309	0.8221	**0.8584**
XuZ	OA	0.5778	0.6147	0.6638	0.6838	0.6111	0.8178	0.7834	0.8235	—	0.9157	0.9192	**0.9765**
	NMI	0.4049	0.5059	0.5357	0.5969	0.6253	0.7919	0.6975	0.6871	—	0.8862	0.8739	**0.9272**
	Kappa	0.4175	0.5155	0.4767	0.6592	0.4361	0.7459	0.6470	0.7620	—	0.8470	0.8649	**0.9681**

of categories, h_{i+} represents the total number of samples of the ith category, h_{+i} represents the number of samples classified as the ith, and N is the total number of class samples.

B. Baseline

In our experiment, we compared our proposed model with some benchmark methods, including two classic clustering algorithms, SC and SSC, as well as several outstanding clustering algorithms developed in recent years.

1) k-means [17] is an outstanding representative of traditional unsupervised learning.
2) CL-CNN is using comparative learning combined with CNN for feature extraction and then clustering the features using k-means.
3) SSC [26] maintains subspace data associations by imposing sparse regularization.
4) l_2-SSC [28] combines the l_2-norm and considers more extensive spatial constraints.
5) EGCSC [36] first introduces GCSC for HSI.
6) RMMF [52] integrates spectral–spatial information into nonnegative matrix factorization for affinity matrix learning.
7) NMFAML [70] integrates the spatial and spectral information of HSI samples into nonnegative matrix factorization for affinity matrix learning.
8) GCOT [53] introduces optimal transport into graph representation learning, facilitating the subsequent learning of natural affinity matrices.
9) GR-RSCNet [35] employs a deep neural network architecture combined with graph regularization for HSI subspace clustering.
10) HMSC [38] constructs multiview data and integrates local and nonlocal spatial information from each view.
11) HGCSC [37] introduces a multihop hypergraph convolution, representing intraclass relationships as edges in the hypergraph.

C. Quantitative Results

Before conducting experiments, we adjust the optimal parameters of the CMSCGC. The specific information is shown in Table II. Table III shows the clustering performance of the CMSCGC and other models for the four datasets. Our model achieved the best clustering performance in most cases across the three evaluation metrics, OA, NMI, and Kappa, significantly outperforming the other methods. Specifically, for the Indian Pines, Houston, and Xu Zhou datasets, the proposed method achieved a 3%–5% improvement in accuracy compared with the second-best approach. Furthermore, we observed the following trends.

1) The introduction of deep learning methods and regularization in clustering effectively improves accuracy. Deep-learning-based approaches, such as GR-RSCNet, show significant improvements compared to traditional models, indicating that nonlinear projections help extract more suitable representations for clustering complex real-world datasets. This ineffectiveness of CL-CNN can be attributed to the utilization of the k-means algorithm for clustering, which is not as accurate as subspace clustering. Furthermore, using CNN for feature extraction fails to capture neighborhood information, thereby diminishing the quality of feature extraction. In traditional models, l_2-SSC introduces l_2-norm regularization, improving the accuracy by 10% on the Indian Pines dataset compared with the SSC method. The EGCSC introduces graph regularization and our model introduces contrastive learning regularization, which also achieves significant improvements.
2) The introduction of complementary information from multiple views improved the clustering accuracy. In addition to our model, NMFAML is another method in the comparison experiments that extracts additional feature information by incorporating homogeneous information. Similarly, our model combines textural information and uses a GCN to aggregate neighborhood information and effectively improve clustering accuracy, achieving 97.61%, 96.69%, 87.21%, and 97.65% accuracy on the Indian Pines, Pavia University, Houston2013, and Xu Zhou datasets, respectively.
3) CMSCGC outperformed EGCSC in terms of the experimental accuracy for all four datasets. Specifically, our model improved by 12.48%, 11.87%, 24.83%, and 19.31% for all datasets. This indicates that the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:41:11 UTC from IEEE Xplore. Restrictions apply.

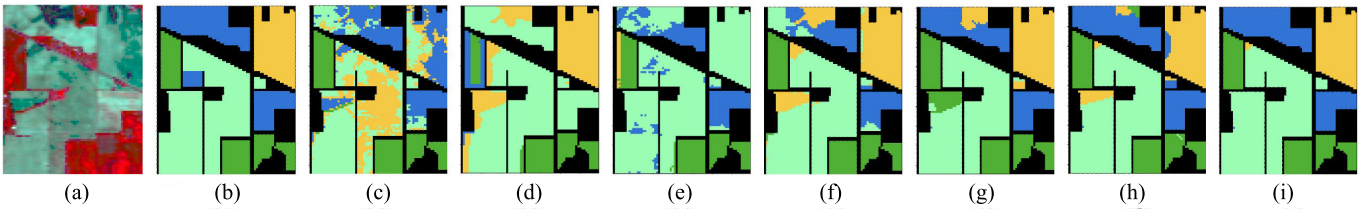


Fig. 2. Cluster visualization on the Indian Pines dataset. (a) False-color image. (b) Ground truth. (c) k-means. (d) SSC. (e) l_2-SSC. (f) EGCSC. (g) HGCSC. (h) GR-RSCNet. (i) Our proposed CMSCGC.

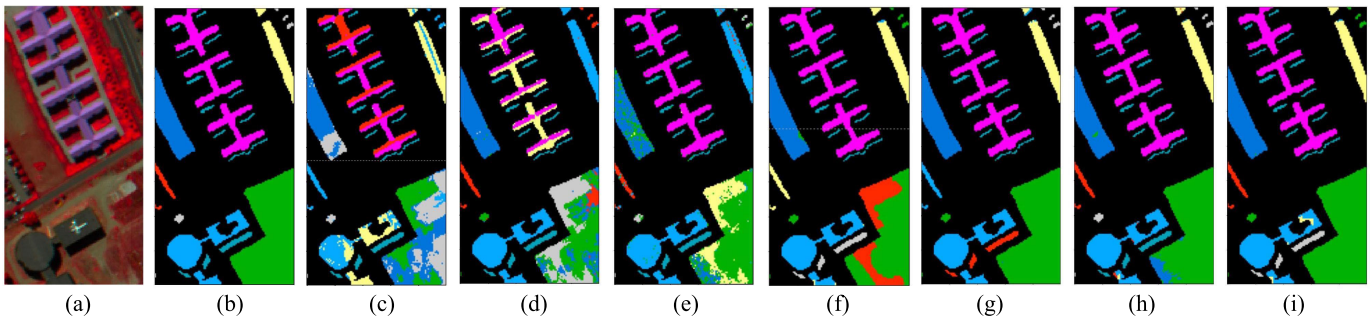


Fig. 3. Cluster visualization on the Pavia University dataset. (a) False-color image. (b) Ground truth. (c) k-means. (d) SSC. (e) l_2-SSC. (f) EGCSC. (g) HGCSC. (h) GR-RSCNet. (i) Our proposed CMSCGC.

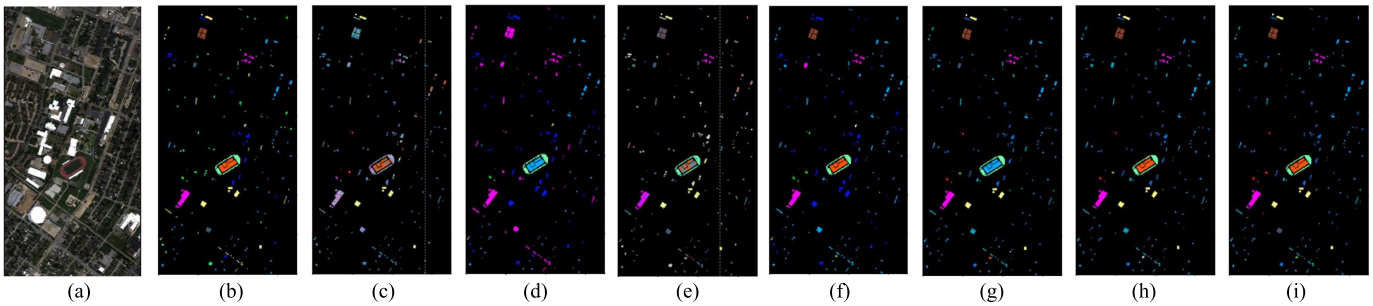


Fig. 4. Cluster visualization on the Houston-2013 dataset. (a) Original image. (b) Ground truth. (c) k-means. (d) SSC. (e) l_2-SSC. (f) EGCSC. (g) HGCSC. (h) GR-RSCNet. (i) Our proposed CMSCGC.

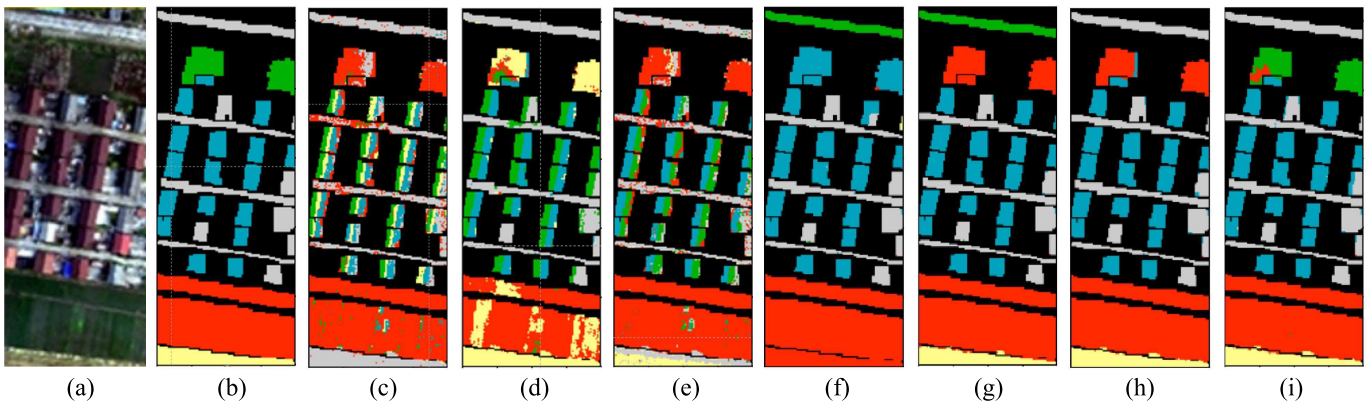


Fig. 5. Cluster visualization on the Xu Zhou dataset. (a) Original image. (b) Ground truth. (c) k-means. (d) SSC. (e) l_2-SSC. (f) EGCSC. (g) HGCSC. (h) GR-RSCNet. (i) Our proposed CMSCGC.

introduction of multiview information, contrastive learning regularization, and attention fusion modules can effectively enhance clustering accuracy. Our model provides a new approach for HSI clustering.

D. Qualitative Comparison of Different Methods

Figs. 2–5 show the visualization results of various clustering methods on the Indian Pines, Pavia University, Houston2013, and Xu Zhou datasets, respectively. Section (a) of each image shows the ground truth of the corresponding dataset with the irrelevant background removed. As spectral clustering is an unsupervised method, the colors representing the same class in different methods may vary; however, this does not affect our visualization results. SSC algorithms can extract low-dimensional information from high-dimensional data structures but do not consider spatial constraints. More-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:41:11 UTC from IEEE Xplore. Restrictions apply.

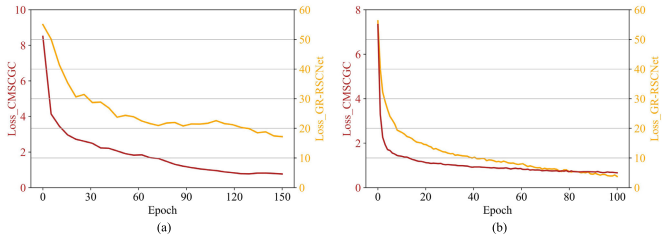


Fig. 6. Loss curves of the CMSCGC and GR-RSCNet. (a) Indian Pines. (b) Pavia University.

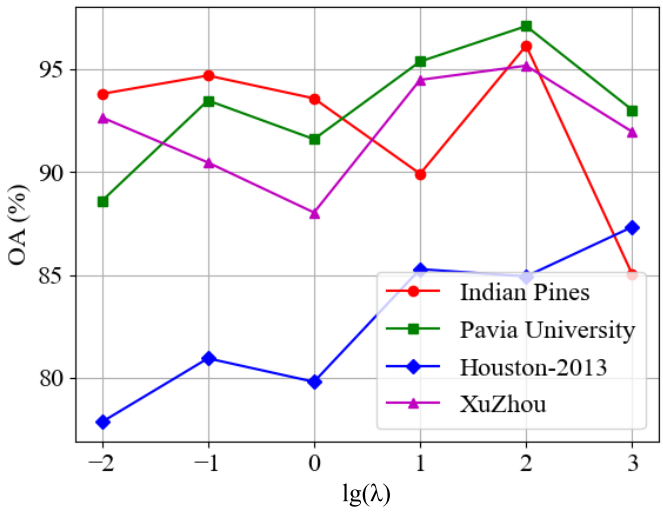


Fig. 7. Performance comparison on four datasets with different regularization coefficients λ.

over, traditional subspace clustering methods struggle to accurately model the HSI structure, making them inferior to deep-learning-based methods.

Our model achieved the best visualization results for all the datasets. Owing to the adaptive aggregation of textural information, the CMSCGC has the least salt-and-pepper noise in the classification images compared to the other methods. Simultaneously, because the GCN model can aggregate neighboring node information, it maintains improved homogeneity within the same class of regions, and the image boundaries remain relatively intact. We also compared the loss curves of the deep learning-based GR-RSCNet method and the proposed CMSCGC method, and the results in Indian Pines and Pavia University are shown in Fig. 6. From the results, we can see that our method not only has better convergence but also can reach convergence faster, which saves the training time of the model to some extent.

E. Parameter Analysis

In this study, we investigated the impacts of the three most important hyperparameters on the CMSCGC framework. Table II provides the set of suitable parameter settings used in the experiments.

1) Sensitivity to Regularization Coefficient λ: We chose λ from the set $S = \{0.01, 0.1, 1, 10, 100, 1000\}$, and Fig. 7 shows the performance of CMSCGC. On the Houston dataset, relatively smaller λ values result in lower OA, NMI, and Kappa values. This is possible because the Houston dataset contains similar dispersed data; therefore, when performing SE, the regularization term to learn the structural affinity matrix is crucial to maintaining connectivity within the same subspace. In the Xu Zhou dataset, our method can achieve nearly optimal performance in terms of the three metrics in a wide range of λ values, indicating that our method has certain robustness to λ.

2) Sensitivity to Nearest Neighbors k: Nearest neighbors k are used to construct the graph embedding of the HSI dataset. The value of k ranges from 20 to 40, with increments of 5. As shown in Fig. 8, if k becomes too large, all three metrics of the model decrease. This is because an excessively large k can lead to the oversmoothing problem. Therefore, we set the k values for the four datasets to 25, 30, 25, and 35.

3) Sensitivity to Patch Window Size w: Window size w is the size of the spatial window used to extract spatial information. We conducted experiments with other fixed parameters to explore the influence of w. The value of w starts at 5, with increments of 2, until reaching 13. As shown in Fig. 9, if w is too small, the patches may contain more redundant spatial information, resulting in a loss of accuracy. Conversely, the model may not be able to extract sufficient spatial information, which would also reduce its performance. When the parameter w is set to 11, 11, 11, and 9, CMSCGC achieves the best results for the four datasets.

F. Ablation Experiments

We conducted various ablation experiments on each module in the network to consider the contributions of the different modules to the rationality of the network structure. Table IV presents the quantitative results of the model on the Indian Pines, Pavia University, Houston2013, and Xu Zhou datasets, with the results being the average of ten experiments.

1) Ablation Experiment for Multiview: Although we have already concluded from the previous results that multiview complementary information can improve clustering accuracy, we demonstrated this more intuitively by comparing the results of a single spectral view, textural view, and multiview (Table IV). The results of a single view are not satisfactory, whereas the fusion of multiview information can significantly improve the accuracy. Although the individual textural view does not surpass the spatial–spectral view in terms of accuracy, the multiview results show improvements of 7.68%, 2.85%, 18.56%, and 10.61% on the Indian Pines, Pavia University, Houston-2013, and Xu Zhou datasets, respectively. This indicates that textural features can complement the missing information in the spatial–spectral view; using textural features, the potential spatial information of land cover can be better exploited. The combination of both leads to an improved clustering accuracy.

2) Ablation Study on Contrastive Learning: Table IV compares the clustering accuracies with and without contrastive learning loss (Cases 3 and 4). We observed that for each dataset, the model with contrastive learning consistently outperformed the simplified model without contrastive learning, achieving higher OA, NMI, and Kappa accuracy. After adding contrastive learning, the model improved by 2.9%, 3.76%,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:41:11 UTC from IEEE Xplore. Restrictions apply.

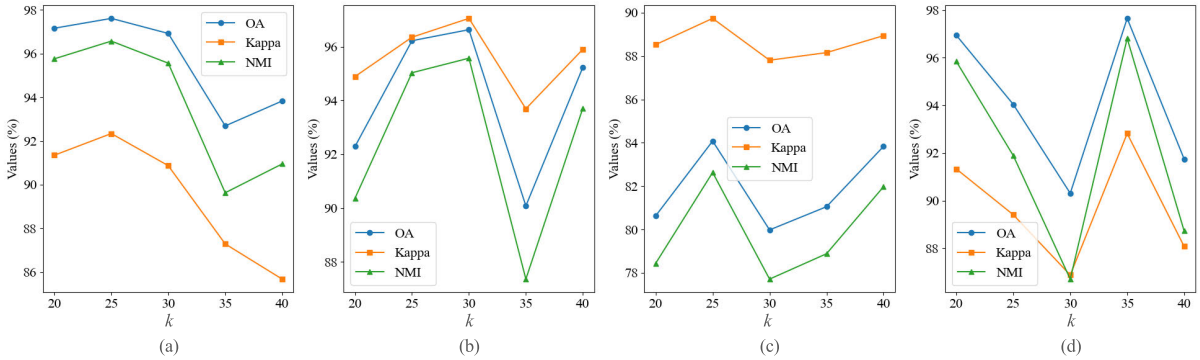


Fig. 8. Performance comparison on four datasets with a different number of neighbors k. (a) Indian Pines dataset. (b) Pavia University dataset. (c) Houston-2013 dataset. (d) Xu Zhou dataset.

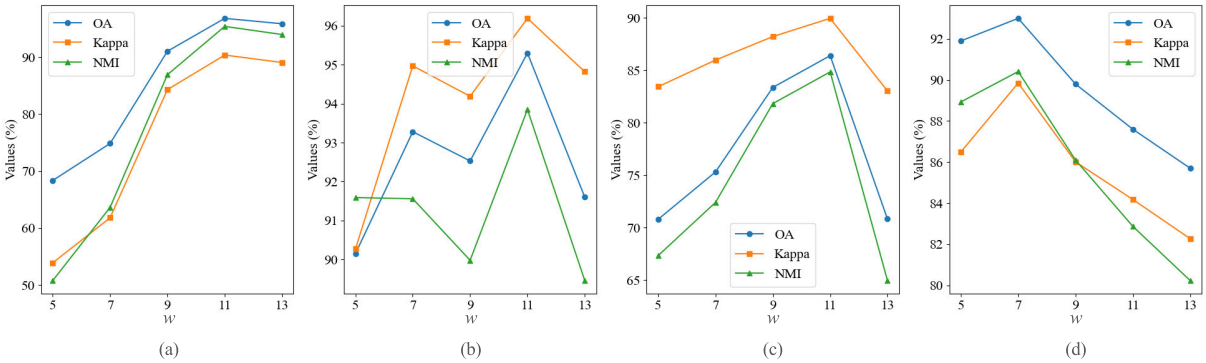


Fig. 9. Performance comparison on four datasets with different input sizes w. (a) Indian Pines dataset. (b) Pavia University dataset. (c) Houston-2013 dataset. (d) Xu Zhou dataset.

TABLE IV

ABLATION STUDIES (OA) FOR DIFFERENT VIEWS, CONTRASTIVE LEARNING, AND ATTENTION FUSION MODULE FOR CMSCGC. THE BEST AND SUBOPTIMAL RESULTS ARE SHOWN IN BOLD AND UNDERLINED, RESPECTIVELY

No.	Texture Features	Spectral-spatial Features	Contrastive Learning	Attention Fusion	InP	PaU	Hou	XuZ
1	✓	×	×	×	0.6961	0.8479	0.6096	0.7678
2	×	✓	×	×	0.8483	0.8442	0.6238	0.7834
3	✓	✓	×	×	0.9251	0.8727	0.8094	0.8895
4	✓	✓	✓	×	<u>0.9541</u>	<u>0.9103</u>	<u>0.8545</u>	<u>0.9504</u>
5	✓	✓	✓	✓	**0.9761**	**0.9669**	**0.8721**	**0.9765**

TABLE V

RUNNING TIME OF DIFFERENT METHODS (IN SECONDS)

Method	InP	PaU	Hou	XuZ
CL-CNN	873.9	1520.1	3534.7	13619.6
SSC	226.5	439.3	907.0	657.3
EGCSC	23.9	53.3	55.5	201.4
GCOT	21.6	64.1	121.4	209.2
HGCSC	54.0	204.9	9353.1	421.3
GR-RSCNet	3282.6	10134.7	14876.0	6077.6
CMSCGC	96.1	614.5	3269.1	754.2

TABLE VI

CLUSTERING PERFORMANCE OF DIFFERENT VIEWS FOR THE INDIAN PINE DATASET

No.	View	OA	NMI	Kappa
1	HSI+EMP	0.9761	0.9234	0.9567
2	HSI+Gabor	0.9604	0.9163	0.9486
3	HSI+Gabor+EMP	0.9832	0.9329	0.9585

4.51%, and 6.09% for the four datasets. This confirms that contrastive loss significantly improves clustering performance.

3) Ablation Study on Adaptive Affinity Matrix: In Table IV, Case 5 represents a simple variant of our model, which only averages the affinity matrices instead of using the attention-based fusion module for spectral clustering. A comparison of Cases 4 and 5 highlights the advantages of the attention-mechanism-based adaptive fusion module. It improved the performance on the four datasets by 2.2%, 5.66%, 1.76%, and 2.61%. This indicates that effectively integrating information from multiview feature spaces can lead to more efficient HSI clustering.

We also conducted a visual comparison of the learned affinity matrices on the Indian Pines dataset; the visualization

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:41:11 UTC from IEEE Xplore. Restrictions apply.

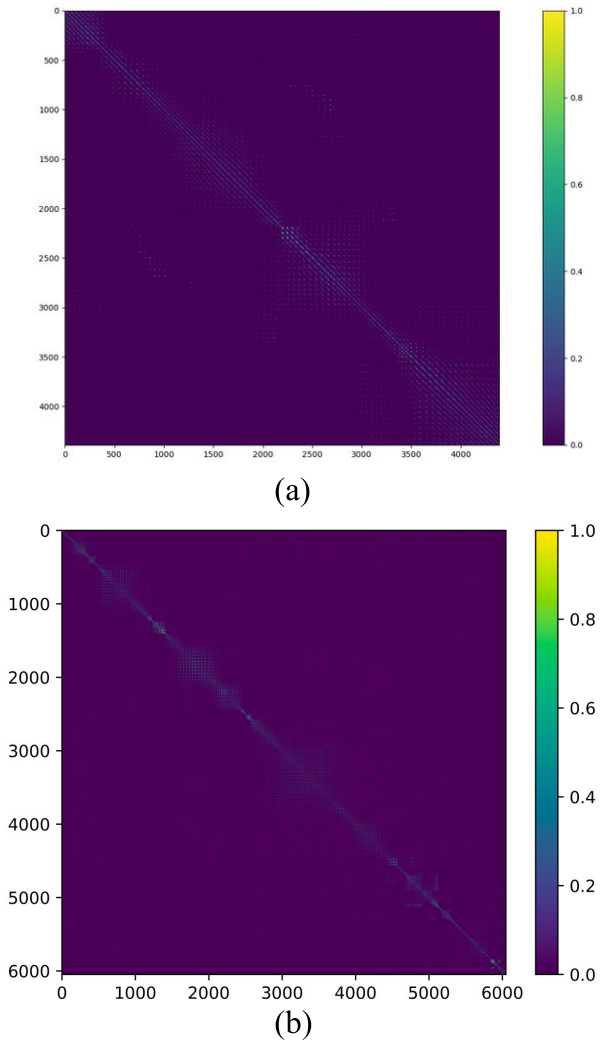


Fig. 10. Visualization of the affinity matrix learn on (a) Indian Pines and (b) Houston.

TABLE VII
CLUSTERING PERFORMANCE OF DIFFERENT LOSS FUNCTIONS FOR THE INDIAN PINE DATASET

Method	InP	PaU	Hou	XuZ
MSCGC	0.9251	0.8727	0.8094	0.8895
CMSCGC w/o NS	0.9317	0.9025	0.7947	0.9036
MSCGC w BT	0.9334	0.9361	0.8054	0.9277
MSCGC w RC	0.9589	0.9434	0.8375	0.9491
CMSCGC	0.9761	0.9669	0.8721	0.9765

results are presented in Fig. 10. As can be seen from the three subfigures, the affinity matrix learned by CMSCGC is not only sparse but also exhibits a more pronounced block diagonal structure, indicating that our model is capable of better identifying intracluster relationships. This can be attributed to the adaptive utilization of textural and spectral information, which leads to accurate HSI classification.

G. Running Time Comparison

Table V presents the runtime performance of various shallow clustering methods (SSC, EGCSC, GCOT, and HGCSC) and deep clustering algorithms (CL-CNN, GR-RSCNet, and CMSCGC). To ensure a fair comparison, all algorithms are executed on the same GPU. The results indicate that shallow clustering methods exhibit faster execution times, as they bypass a training process and directly perform clustering after obtaining shallow features. On the other hand, deep clustering algorithms, such as CL-CNN, GR-SCNet, and CMSCGC, require multiple rounds of operations during training to acquire more robust features. Notably, despite both being deep clustering algorithms, CMSCGC outperforms GR-SCNet in terms of speed. This observation underscores that our proposed GR-RSCNet method achieves a commendable balance between computational cost and clustering performance.

H. Clustering With Different Views

To illustrate the application of CMSCGC in the multiview domain, we present the clustering results on the Indian Pines dataset using different views in Table VI. In the table, "HSI" denotes the original features of the image, i.e., spectral spatial features. "EMP" and "Gabor" represent the view information used for feature extraction, employing EMP and 2-D Gabor filters, respectively. The results indicate that the model inputting three views of information achieves a 0.71% and 2.28% higher OA compared to the OA obtained by inputting two views. This not only demonstrates that our model adeptly handles multiview information but also underscores the complementary role that multiview information can play in enhancing clustering tasks.

I. Clustering With Different Loss Functions

Various prevailing clustering methods fall into two primary categories: contrastive clustering and reconstructive clustering [71]. Among contrastive-based methods, they are further distinguished by the presence or absence of negative samples [58]. We employ different loss functions to compare clustering results, namely, lossless multiview subspace clustering based on graph convolutional networks (MSCGC), contrastive clustering without negative samples (CMSCGC w/o NS), reconstructive clustering [72] (MSCGC w RC), and standard contrastive clustering (CMSCGC). All the aforementioned approaches involve individual sample-level contrastive learning. In addition, we construct overall feature-level comparative methods based on the original MSCGC model and the Barlow Twins loss [60] (MSCGC w BT). As observed in the results presented in Table VII, our proposed CMSCGC demonstrates optimal performance, while reconstructive clustering yields suboptimal results. This discrepancy arises from the fact that MSCGC lacks a feature optimization process, focusing solely on extracting shallow features. MSCGC with BT lacks interclass discrimination, thereby resulting in unsatisfactory outcomes. On the other hand, CMSCGC without NS diminishes the clustering effect due to the absence of negative samples, preventing precise separation of distances between clusters.

V. CONCLUSION

In this article, we develop a novel multiview subspace clustering network called CMSCGC for HSI clustering.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:41:11 UTC from IEEE Xplore. Restrictions apply.

Specifically, we first extract textural and spectral information from HSIs to construct a multiview graph structure. We then use GCNs to aggregate neighborhood information and employ contrastive learning to learn the consistency and complementary information of the multiview data. The extracted robust features are then fed into a self-expression network to learn the affinity matrices. Finally, considering the varying contributions of different views to the clustering task, we adopt an attention-based strategy to combine the learning of the two branches and obtain more discriminative affinity matrices. We test our proposed model on four common hyperspectral datasets and compared it with state-of-the-art models. The experimental results show that CMSCGC achieved the best clustering performance. Comprehensive ablation studies are also conducted to verify the effectiveness of the designed modules.

The self-expression layer makes it difficult for the proposed model to be trained on large-scale datasets. Although the min-batch method can alleviate computational pressure, it is challenging to fully exploit the capabilities of GCNs. In the future, we will attempt to explore the potential of CMSCGC using large-scale datasets.

ACKNOWLEDGMENT

Computation of this study was performed by the High-Performance GPU Server (TX321203) Computing Centre of the National Education Field Equipment Renewal and Renovation Loan Financial Subsidy Project, China University of Geosciences, Wuhan, China.

REFERENCES

[1] G. Camps-Valls, D. Tuia, L. Bruzzone, and J. A. Benediktsson, "Advances in hyperspectral image classification: Earth monitoring with statistical learning methods," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 45–54, Jan. 2014.

[2] Y.-P. Zhao, H. Li, Y. Chen, Z. Wang, and X. Li, "Hyperspectral anomaly detection via structured sparsity plus enhanced low-rankness," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–15, 2023, Art. no. 5515115.

[3] R. Guan, Z. Li, T. Li, X. Li, J. Yang, and W. Chen, "Classification of heterogeneous mining areas based on ResCapsNet and GaoFen-5 imagery," *Remote Sens.*, vol. 14, no. 13, p. 3216, Jul. 2022.

[4] Y. Chen, Q. Yuan, Y. Tang, Y. Xiao, J. He, and L. Zhang, "SPIRIT: Spectral awareness interaction network with dynamic template for hyperspectral object tracking," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–16, 2024, Art. no. 5503116.

[5] Y. Tang et al., "An object fine-grained change detection method based on frequency decoupling interaction for high-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–13, 2024, Art. no. 5600213.

[6] C. Zhao et al., "Hyperspectral image classification with multi-attention transformer and adaptive superpixel segmentation-based active learning," *IEEE Trans. Image Process.*, vol. 32, pp. 3606–3621, 2023.

[7] Y. Chen, Y. Tang, Z. Yin, T. Han, B. Zou, and H. Feng, "Single object tracking in satellite videos: A correlation filter-based dual-flow tracker," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 6687–6698, 2022.

[8] W. Chen, S. Ouyang, J. Yang, X. Li, G. Zhou, and L. Wang, "JAGAN: A framework for complex land cover classification using GaoFen-5 AHSI images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1591–1603, 2022.

[9] J. Liu, R. Guan, Z. Li, J. Zhang, Y. Hu, and X. Wang, "Adaptive multi-feature fusion graph convolutional network for hyperspectral image classification," *Remote Sens.*, vol. 15, no. 23, p. 5483, Nov. 2023.

[10] Y. Tarabalka, M. Fauvel, J. Chanussot, and J. A. Benediktsson, "SVM- and MRF-based method for accurate classification of hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 7, no. 4, pp. 736–740, Oct. 2010.

[11] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.

[12] R. Wang, F. Nie, and W. Yu, "Fast spectral clustering with anchor graph for large hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 11, pp. 2003–2007, Nov. 2017.

[13] H. Zhai, H. Zhang, P. Li, and L. Zhang, "Hyperspectral image clustering: Current achievements and future lines," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 4, pp. 35–67, Dec. 2021.

[14] D. Hu, K. Liang, S. Zhou, W. Tu, M. Liu, and X. Liu, "ScDFC: A deep fusion clustering method for single-cell RNA-seq data," *Briefings Bioinf.*, vol. 24, no. 4, Jul. 2023, Art. no. bbad216.

[15] R. A. Borsoi et al., "Spectral variability in hyperspectral data unmixing: A comprehensive review," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 4, pp. 223–270, Dec. 2021.

[16] J. Chang, G. Meng, L. Wang, S. Xiang, and C. Pan, "Deep self-evolution clustering," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 4, pp. 809–823, Apr. 2020.

[17] T. Kanungo, D. M. Mount, N. S. Netanyahu, C. D. Piatko, R. Silverman, and A. Y. Wu, "An efficient k-means clustering algorithm: Analysis and implementation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 24, no. 7, pp. 881–892, Jul. 2002.

[18] S. Ghaffarian and S. Ghaffarian, "Automatic histogram-based fuzzy C-means clustering for remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 97, pp. 46–57, Nov. 2014.

[19] D. Comaniciu and P. Meer, "Mean shift: A robust approach toward feature space analysis," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 24, no. 5, pp. 603–619, May 2002.

[20] Y. Chen, S. Ma, X. Chen, and P. Ghamisi, "Hyperspectral data clustering based on density analysis ensemble," *Remote Sens. Lett.*, vol. 8, no. 2, pp. 194–203, Feb. 2017.

[21] R. Vidal, "Subspace clustering," *IEEE Signal Process. Mag.*, vol. 28, no. 2, pp. 52–68, Mar. 2011.

[22] S. Liu et al., "Efficient one-pass multi-view subspace clustering with consensus anchors," in *Proc. AAAI Conf. Artif. Intell.*, 2022, vol. 36, no. 7, pp. 7576–7584.

[23] S. Liu, X. Liu, S. Wang, X. Niu, and E. Zhu, "Fast incomplete multi-view clustering with view-independent anchors," *IEEE Trans. Neural Netw. Learn. Syst.*, 2022, doi: 10.1109/TNNLS.2022.3220486.

[24] Q. Ou, S. Wang, S. Zhou, M. Li, X. Guo, and E. Zhu, "Anchor-based multiview subspace clustering with diversity regularization," *IEEE MultimediaMag.*, vol. 27, no. 4, pp. 91–101, Oct. 2020.

[25] Q. Li, W. Liu, and L. Li, "Affinity learning via a diffusion process for subspace clustering," *Pattern Recognit.*, vol. 84, pp. 39–50, Dec. 2018.

[26] H. Zhang, H. Zhai, L. Zhang, and P. Li, "Spectral–spatial sparse subspace clustering for hyperspectral remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 6, pp. 3672–3684, Jun. 2016.

[27] R. Vidal and P. Favaro, "Low rank subspace clustering (LRSC)," *Pattern Recognit. Lett.*, vol. 43, pp. 47–61, Jul. 2014.

[28] H. Zhai, H. Zhang, L. Zhang, P. Li, and A. Plaza, "A new sparse subspace clustering algorithm for hyperspectral remote sensing imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 1, pp. 43–47, Jan. 2017.

[29] S. Huang, H. Zhang, and A. Pižurica, "Semisupervised sparse subspace clustering method with a joint sparsity constraint for hyperspectral remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 3, pp. 989–999, Mar. 2019.

[30] W. Tu et al., "Attribute-missing graph clustering network," in *Proc. 38th AAAI Conf. Artif. Intell. (AAAI)*, 2024.

[31] Z. Peng, H. Liu, Y. Jia, and J. Hou, "Adaptive attribute and structure subspace clustering network," *IEEE Trans. Image Process.*, vol. 31, pp. 3430–3439, 2022.

[32] M. Liu, Y. Liu, K. Liang, S. Wang, S. Zhou, and X. Liu, "Deep temporal graph clustering," 2023, *arXiv:2305.10738*.

[33] J. Lei, X. Li, B. Peng, L. Fang, N. Ling, and Q. Huang, "Deep spatial–spectral subspace clustering for hyperspectral image," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 31, no. 7, pp. 2686–2697, Jul. 2021.

[34] K. Li, Y. Qin, Q. Ling, Y. Wang, Z. Lin, and W. An, "Self-supervised deep subspace clustering for hyperspectral images with adaptive self-expressive coefficient matrix initialization," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 3215–3227, 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:41:11 UTC from IEEE Xplore. Restrictions apply.

[35] Y. Cai, M. Zeng, Z. Cai, X. Liu, and Z. Zhang, "Graph regularized residual subspace clustering network for hyperspectral image clustering," *Inf. Sci.*, vol. 578, pp. 85–101, Nov. 2021.
[36] Y. Cai, Z. Zhang, Z. Cai, X. Liu, X. Jiang, and Q. Yan, "Graph convolutional subspace clustering: A robust subspace clustering framework for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4191–4202, May 2021.
[37] Z. Zhang, Y. Cai, W. Gong, P. Ghamisi, X. Liu, and R. Gloaguen, "Hypergraph convolutional subspace clustering with multihop aggregation for hyperspectral image," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 676–686, 2022.
[38] S. Huang, H. Zhang, and A. Pizurica, "Hybrid-hypergraph regularized multiview subspace clustering for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022, Art. no. 5505816.
[39] C. Hinojosa, F. Rojas, S. Castillo, and H. Arguello, "Hyperspectral image segmentation using 3D regularized subspace clustering model," *J. Appl. Remote Sens.*, vol. 15, no. 1, Jan. 2021, Art. no. 016508.
[40] J. A. Benediktsson, J. A. Palmason, and J. R. Sveinsson, "Classification of hyperspectral data from urban areas based on extended morphological profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 480–491, Mar. 2005.
[41] W. Li, C. Chen, H. Su, and Q. Du, "Local binary patterns and extreme learning machine for hyperspectral imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 7, pp. 3681–3693, Jul. 2015.
[42] J. Wang, C. Tang, Z. Wan, W. Zhang, K. Sun, and A. Y. Zomaya, "Efficient and effective one-step multiview clustering," *IEEE Trans. Neural Netw. Learn. Syst.*, 2023, doi: 10.1109/TNNLS.2023.3253246.
[43] L. Tian, Q. Du, I. Kopriva, and N. Younan, "Spatial–spectral based multiview low-rank sparse sbuspace clustering for hyperspectral imagery," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2018, pp. 8488–8491.
[44] Z. Chen, C. Zhang, T. Mu, and Y. He, "Tensorial multiview subspace clustering for polarimetric hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022, Art. no. 5528513, doi: 10.1109/TGRS.2022.3174830.
[45] H. Lu, H. Su, J. Hu, and Q. Du, "Dynamic ensemble learning with multiview kernel collaborative subspace clustering for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 2681–2695, 2022.
[46] Y. Cai et al., "Superpixel contracted neighborhood contrastive subspace clustering network for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022, Art. no. 5530113.
[47] Y. Zhu, Y. Xu, F. Yu, Q. Liu, S. Wu, and L. Wang, "Graph contrastive learning with adaptive augmentation," in *Proc. Web Conf.*, Apr. 2021, pp. 2069–2080.
[48] Y. Xie, W. Zhang, Y. Qu, L. Dai, and D. Tao, "Hyper-Laplacian regularized multilinear multiview self-representations for clustering and semisupervised learning," *IEEE Trans. Cybern.*, vol. 50, no. 2, pp. 572–586, Feb. 2020.
[49] D. Hu, Z. Dong, K. Liang, J. Wang, S. Wang, and X. Liu, "Single-cell multi-view clustering via community detection with unknown number of clusters," 2023, *arXiv:2311.17103*.
[50] Y. Chen, S. Wang, C. Peng, Z. Hua, and Y. Zhou, "Generalized nonconvex low-rank tensor approximation for multi-view subspace clustering," *IEEE Trans. Image Process.*, vol. 30, pp. 4022–4035, 2021.
[51] Y. Chen, X. Xiao, C. Peng, G. Lu, and Y. Zhou, "Low-rank tensor graph learning for multi-view subspace clustering," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 1, pp. 92–104, Jan. 2021.
[52] L. Zhang, L. Zhang, B. Du, J. You, and D. Tao, "Hyperspectral image unsupervised classification by robust manifold matrix factorization," *Inf. Sci.*, vol. 485, pp. 154–169, Jun. 2019.
[53] S. Liu and H. Wang, "Graph convolutional optimal transport for hyperspectral image spectral clustering," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022, Art. no. 4414013.
[54] R. Liu, "Understand and improve contrastive learning methods for visual representation: A review," 2021, *arXiv:2106.03259*.
[55] W. V. Gansbeke, S. Vandenhende, S. Georgoulis, M. Proesmans, and L. V. Gool, "SCAN: Learning to classify images without labels," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 268–285.
[56] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 1597–1607.
[57] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9729–9738.
[58] J. Grill et al., "Bootstrap your own latent—A new approach to self-supervised learning," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2020, pp. 21271–21284.
[59] X. Chen and K. He, "Exploring simple Siamese representation learning," in *Proc. IEEE Comput. Soc. Conf. Comput. Vision Pattern Recognit.*, Jun. 2021, pp. 15750–15758.
[60] J. Zbontar, L. Jing, I. Misra, Y. LeCun, and S. Deny, "Barlow twins: Self-supervised learning via redundancy reduction," in *Proc. Int. Conf. Mach. Learn. (ICML)*, vol. 139, 2021, pp. 12310–12320.
[61] Y. Liu et al., "Deep graph clustering via dual correlation reduction," in *Proc. AAAI Conf. Artif. Intell.*, vol. 36, no. 7, 2022, pp. 7603–7611.
[62] W. Tu, S. Zhou, X. Liu, C. Ge, Z. Cai, and Y. Liu, "Hierarchically contrastive hard sample mining for graph self-supervised pretraining," *IEEE Trans. Neural Netw. Learn. Syst.*, 2024, doi: 10.1109/TNNLS.2023.3297607.
[63] R. Guan, Z. Li, X. Li, and C. Tang, "Pixel-superpixel contrastive learning and pseudo-label correction for hyperspectral image clustering," 2023, *arXiv:2312.09630*.
[64] S. Hou, H. Shi, X. Cao, X. Zhang, and L. Jiao, "Hyperspectral imagery classification based on contrastive learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022, Art. no. 5521213.
[65] J.-X. Wang, T. Li, S.-B. Chen, J. Tang, B. Luo, and R. C. Wilson, "Reliable contrastive learning for semi-supervised change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022, Art. no. 4416413.
[66] H. Li et al., "Global and local contrastive self-supervised learning for semantic segmentation of HR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022, Art. no. 5618014.
[67] M. Wang, F. Gao, J. Dong, H.-C. Li, and Q. Du, "Nearest neighbor-based contrastive learning for hyperspectral and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–16, 2023, Art. no. 5501816.
[68] C. Rodarmel and J. Shan, "Principal component analysis for hyperspectral image classification," *Surv. Land Inf. Syst.*, vol. 62, no. 2, pp. 115–122, Jun. 2002.
[69] M. Fauvel, Y. Tarabalka, J. A. Benediktsson, J. Chanussot, and J. C. Tilton, "Advances in spectral–spatial classification of hyperspectral images," *Proc. IEEE*, vol. 101, no. 3, pp. 652–675, Mar. 2013.
[70] Y. Qin, B. Li, W. Ni, S. Quan, P. Wang, and H. Bian, "Affinity matrix learning via nonnegative matrix factorization for hyperspectral imagery clustering," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 402–415, 2021.
[71] Y. Liu et al., "A survey of deep graph clustering: Taxonomy, challenge, application, and open resource," 2022, *arXiv:2211.12875*.
[72] F. Tian, B. Gao, Q. Cui, E. Chen, and T.-Y. Liu, "Learning deep representations for graph clustering," in *Proc. AAAI Conf. Artif. Intell.*, 2014, vol. 28, no. 1, pp. 1293–1299.



Renxiang Guan received the B.S. degree from China University of Geosciences (CUG), Wuhan, China, in 2023. He is currently pursuing the master's degree with the National University of Dense Technology (NUDT), Changsha, China.

He has published several peer-reviewed papers, including AAAI, ICASSP, *Knowledge-Based Systems* (KBS), and *Remote Sensing*. His research interests include deep graph clustering and hyperspectral image processing.



Zihao Li received the B.S. degree from China University of Geosciences (CUG), Wuhan, China, in 2023, where he is currently pursuing the Ph.D. degree.

He has published several peer-reviewed papers, including the International Conference on Acoustics, Speech and Signal Processing (ICASSP), and *Remote Sensing*. His research interests include remote sensing image process, computer vision, and deep learning.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:41:11 UTC from IEEE Xplore. Restrictions apply.



Wenxuan Tu received the Ph.D. degree from the College of Computer, National University of Defense Technology (NUDT), Changsha, China, in 2023.

He has published several papers in highly regarded journals and conferences, such as AAAI Conference on Artificial Intelligence, International Conference on Machine Learning (ICML), IEEE TRANSACTIONS ON MULTIMEDIA (TMM), and IEEE TRANSACTIONS ON IMAGE PROCESSING (TIP).



Jun Wang received the B.E. degree from Hubei University of Technology, Wuhan, China, in 2019, and the M.S. degree from China University of Geosciences, Wuhan, in 2023. He is currently pursuing the Ph.D. degree with the National University of Defense Technology, Changsha, China.

His research interests include hyperspectral image processing and machine learning.



Yue Liu received the M.S. degree from the National University of Dense Technology (NUDT), Changsha, China, in 2023.

He has published several peer-reviewed papers, including the International Conference on Learning Representations (ICLR), AAAI Conference on Artificial Intelligence, the International Joint Conference on Artificial Intelligence (IJCAI), and the ACM International Conference on Multimedia (ACM MM). His research interests include self-supervised learning, knowledge graph, and deep graph clustering.



Xianju Li (Member, IEEE) received the B.S. degree in geomatics engineering, the M.S. degree in geodesy and survey engineering, and the Ph.D. degree in surveying and mapping from China University of Geosciences, Wuhan, China, in 2009, 2012, and 2016, respectively.

Since 2016, he has been an Associate Professor with the School of Computer Science, China University of Geosciences. He has authored and coauthored more than 20 articles. His main research interests include remote sensing image processing and analysis, computer vision, and machine learning.



Chang Tang (Senior Member, IEEE) received the Ph.D. degree from Tianjin University, Tianjin, China, in 2016.

He was with AMRL, University of Wollongong, Wollongong, NSW, Australia, from September 2014 to September 2015. He is currently a Full Professor with the School of Computer Science, China University of Geosciences, Wuhan, China. He has published more than 50 peer-reviewed papers, including those in highly regarded journals and conferences.



Ruyi Feng (Member, IEEE) received the B.S. degree in geographic information system from Hunan Normal University, Changsha, China, in 2011, and the M.S. degree in surveying and mapping engineering and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2013 and 2016, respectively.

Since 2016, she has been with the School of Computer Science, China University of Geosciences, Wuhan, where she is currently an Associate Professor.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:41:11 UTC from IEEE Xplore. Restrictions apply.